

HW 6.2.2: Parametric Functions

Match each set of equations with one of the graphs below.

1. $\begin{cases} x(t) = t \\ y(t) = t^2 - 1 \end{cases}$

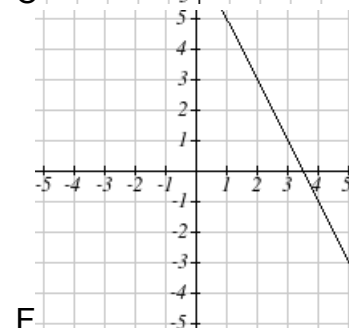
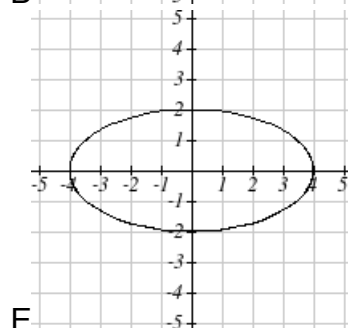
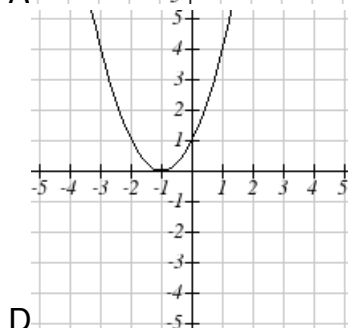
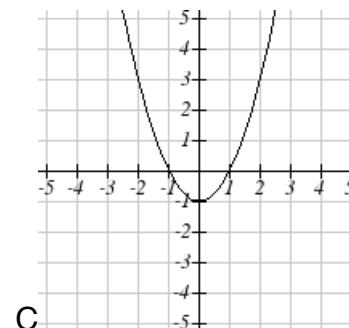
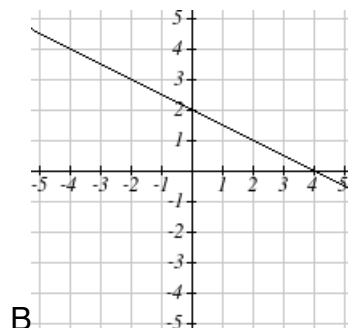
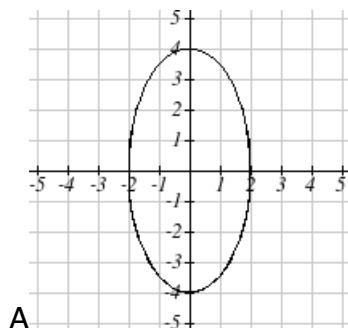
2. $\begin{cases} x(t) = t - 1 \\ y(t) = t^2 \end{cases}$

3. $\begin{cases} x(t) = 4\sin(t) \\ y(t) = 2\cos(t) \end{cases}$

4. $\begin{cases} x(t) = 2\sin(t) \\ y(t) = 4\cos(t) \end{cases}$

5. $\begin{cases} x(t) = 2 + t \\ y(t) = 3 - 2t \end{cases}$

6. $\begin{cases} x(t) = -2 - 2t \\ y(t) = 3 + t \end{cases}$



Sketch the parametric equations for $-2 \leq t \leq 2$.

7. $\begin{cases} x(t) = t^2 + 2 \\ y(t) = 1 + t \end{cases}$

8. $\begin{cases} x(t) = 5 - t^2 \\ y(t) = 2t^2 + 1 \end{cases}$

9. $\begin{cases} x(t) = 1 + 2t \\ y(t) = t^2 \end{cases}$

10. $\begin{cases} x(t) = 2t - 2 \\ y(t) = t^3 \end{cases}$

Sketch the parametric equations

11.
$$\begin{cases} x(t) = 5 \cos(t) \\ y(t) = 2 \sin(t) \end{cases}$$

12.
$$\begin{cases} x(t) = 1 + 2 \cos(t) \\ y(t) = 3 - \sin(t) \end{cases}$$

Eliminate the parameter t to rewrite the parametric equation as a Cartesian equation

13.
$$\begin{cases} x(t) = 5 - t \\ y(t) = 8 - 2t \end{cases}$$

14.
$$\begin{cases} x(t) = 7 + 2t \\ y(t) = 11 - t \end{cases}$$

15.
$$\begin{cases} x(t) = 2t + 1 \\ y(t) = 3\sqrt{t} \end{cases}$$

16.
$$\begin{cases} x(t) = 3t - 1 \\ y(t) = 2t^2 \end{cases}$$

17.
$$\begin{cases} x(t) = 2e^t \\ y(t) = 1 - 5t \end{cases}$$

18.
$$\begin{cases} x(t) = 4 + 2t \\ y(t) = 3 \log(t) \end{cases}$$

19.
$$\begin{cases} x(t) = t^3 - t \\ y(t) = 2t \end{cases}$$

20.
$$\begin{cases} x(t) = t + 1 \\ y(t) = t - t^5 \end{cases}$$

21.
$$\begin{cases} x(t) = e^{2t} \\ y(t) = e^{6t} \end{cases}$$

22.
$$\begin{cases} x(t) = t^3 \\ y(t) = t^9 \end{cases}$$

23.
$$\begin{cases} x(t) = 4 \cos(t) \\ y(t) = 5 \sin(t) \end{cases}$$

24.
$$\begin{cases} x(t) = 4 \sin(t) + 1 \\ y(t) = 7 \cos(t) - 2 \end{cases}$$

Create a parametric equation for the rectangular equations

25.
$$\frac{(x-2)^2}{36} + \frac{(y+3)^2}{25} = 1$$

26.
$$(x+3)^2 + (y-1)^2 = 49$$

27.
$$\frac{(x+4)^2}{16} - \frac{(y-2)^2}{36} = 1$$

28.
$$\frac{(y-5)^2}{81} - \frac{(x+4)^2}{64} = 1$$

Selected Answers:

1. C

3. E

5. F

13. From the first equation, $t = 5 - x$. Substituting into the second equation, $y = 8 - 2(5 - x) = 8 - 10 + 2x$, so the Cartesian equation is $y = 2x - 2$.

15. From the first equation, $t = \frac{x-1}{2}$. Substituting into the second equation, $y = 3\sqrt{\left(\frac{x-1}{2}\right)}$.

17. From the first equation, $t = \ln\left(\frac{x}{2}\right)$. Substituting into the second equation, $y = 1 - 5\ln\left(\frac{x}{2}\right)$.

19. From the second equation, $t = \frac{y}{2}$. Substituting into the first equation, $x = \left(\frac{y}{2}\right)^3 - \frac{y}{2}$.

21. Note that the second equation can be written as $y = (e^{2t})^3$. Then substituting $x = e^{2t}$ from the first equation into this new equation, we get $y = x^3$.

23. From the first equation, $\cos(t) = \frac{x}{4}$. From the second equation, $\sin(t) = \frac{y}{5}$. Since $\sin^2(t) + \cos^2(t) = 1$, we get $\frac{x^2}{4^2} + \frac{y^2}{5^2} = 1$.

$$25. \begin{cases} x(t) = 6\sin(t) + 2 \\ y(t) = 5\cos(t) - 3 \end{cases} \quad (\text{answers may vary})$$

$$27. \begin{cases} x(t) = 4\sec(t) - 4 \\ y(t) = 6\tan(t) + 2 \end{cases} \quad (\text{answers may vary})$$